



UNIVERSITI MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

CSE3433
Software Architecture

Cloud-Native Learning Management System (LMS)
(Cloud-Native / Layered Architecture)

Prepared by:
Group No.5

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COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOUR
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WEEKLY LOG ENTRIES

PHASE 1: Project Proposal

WEEK 5

Undertaken Tasks & Weekly Activities	● Participated in group discussions to identify problems that faced by traditional LMS platforms. ● Chooosed the role for each member. ● Discussed some suitable technologies and database solutions for the proposed cloud-native LMS. ● Assisted in preparing the Project Proposal document for Deliverable 1.
Individual Contributions (as Backend & Database Engineer)	● Suggested the use of RESTful APIs to enable communication between system modules because Cloud-Native LMS consists of multiple independent services. ● Proposed suitable databases such as MongoDB and SQLite for managing LMS data. ● Identified the data requirements for user management, course management, assignment and progress tracking modules. ● Shared ideas on how backend services would interact with the database layer. ● Shared ideas on backend requirements for user authentication and assignment submission modules.
Challenges Encountered	Limit understanding of how database per service concepts work in cloud-native architecture as of how backend services interact with cloud-native environment.
Action Plans	Studied and reviewed some examples of cloud-native LMS implementation and REST API architecture to enhance the understanding.

Table 1 : Weekly Progress Log (Week5)

PHASE 2: Architecture Design

WEEK 6

Undertaken Tasks & Weekly Activities	● Participated in group discussion to analyse the function and non-function requirement of the Cloud-Native LMS. ● Reviewed the proposed architecture style and identified the responsibilities of backend services within the Application Layer. ● Discussed how the frontend, backend services, API Gateway and database would communicate with each other. ● Explore some suitable approaches for implementing RESTful APIs to support LMS functionalities such as user management, assignment submissions and progress tracking.
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	- Identified the data requirement needed by each module and how these modules would interact with database layer. - Assisted in refining the system overview and ensuring that backend-related components aligned with the proposed architecture design.
Individual Contributions (as Backend & Database Engineer)	- Identified backend responsibilities for each microservice or layer including User Management, Assignment Service and Notification Service. - Discussed the finding about the database integration requirement for the proposed system by identifying how backend services would connect and interact with the database layer. - Suggested using RESTful APIs to provide standardized communication between system components. - Reviewed the proposed service interactions to ensure the backend design supported scalability and maintainability.
Challenges Encountered	Faced difficulties in determining clear responsibilities for each microservice while ensuring effective communication between services. It was also challenging to balance system scalability with maintainability during the database and backend design process.
Action Plans	Will refine the database design and backend workflows based on feedback while enhancing knowledge of RESTful APIs and backend technologies. API interactions will also be documented to support future prototype implementation.

Table 2 : Weekly Progress Log (Week 6)

WEEK 7

Undertaken Tasks & Weekly Activities	- Conducted a comprehensive review of the finalized architectural diagrams and technical documentation prior to the Deliverable 2 submission. - Engaged in group consultative sessions to validate the alignment of the proposed backend framework with established system specifications. - Verified the visual representation of workflows between frontend modules, backend logic, and the persistent storage layer within the diagrams. - Collaborated with team members to detect and rectify any technical discrepancies or omissions in the interaction and deployment models. - Participated in the final assessment of the architecture report to confirm that every service facilitates the core functionalities of the LMS.
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Individual Contributions (as Backend Developer)	● Reviewed the sequence diagrams involving backend processes such as user registration, assignment submission, notifications, and progress tracking. ● Verified that the proposed backend workflows supported the functional requirements specified in the report. ● Ensured that the responsibilities assigned to each microservice were clearly defined without overlapping functionalities. ● Examined whether database interactions illustrated in the diagrams reflected the intended flow of data between the application and data layers. ● Assisted in proofreading backend-related sections of the report to maintain consistency and technical accuracy.
Challenges Encountered	It was challenging to ensure that the sequence diagrams accurately represented the backend workflows while remaining clear and easy to understand. Maintaining consistency between the functional requirements and service interactions also required several revisions.
Action Plans	Plan to improve the accuracy of the architecture documentation by reviewing service interactions and backend workflows. Further preparation for prototype implementation will be carried out through backend and database integration studies.

Table 3 : Weekly Progress Log (Week 7)

PHASE 3: Project Observation 1 & Prototype Progress

WEEK 8

Undertaken Tasks & Weekly Activities	● Participated in group discussions to prepare for Project Observation 1. ● Evaluated the architectural visualisations and designated essential backend modules for initial focus in the upcoming prototyping stage. ● Started planning the implementation approach for backend services required by the LMS. ● Discussed database structures and the data flow needed to support the core functionalities of the system. ● Prepared references and notes related to backend development to facilitate the transition from design to implementation.
Individual Contributions (as Backend Developer)	● Identified backend functionalities that should be implemented first such as user registration, login authentication and assignment submission. ● Planned the database tables required to support the prototype, including user, course, assignment, notification and progress tracking records.

	● Drafted the expected CRUD operations that each backend service would perform. ● Reviewed how RESTful APIs could be structured to support communication between the frontend and backend components. ● Assisted the team in preparing explanations regarding backend processes for Project Observation 1.
Challenges Encountered	Encountered difficulties in prioritising backend features and translating the proposed architecture into a practical, scalable prototype design.
Action Plans	Will prioritise essential backend functionalities and refine the database design to support the upcoming prototype development. Feedback from Project Observation 1 will also be incorporated to improve the implementation approach and backend planning.

Table 4 : Weekly Progress Log (Week 8)

Tool Used	ChatGPT
Purpose	Assisting in planning the implementation of backend services for the LMS prototype.
Prompt Used	"What backend functionalities should be prioritised when developing the first prototype of a cloud-native learning management system?"
Output Summary	Suggested prioritising core functionalities such as user authentication, course management, assignment submission, and database integration before implementing advanced features.
How It Was Used	The recommendations were used as a guideline to determine the priority of backend functionalities during the early prototype stage. Based on the suggestions, I focused on planning essential components such as user authentication, assignment submission, and database integration before considering additional features. This helped the team prepare a clearer explanation of the prototype scope and backend workflow for Project Observation 1.

Table 5 : GenAI Usage Record (Week 8)

WEEK 9

Undertaken Tasks & Weekly Activities	● Participated in Project Observation 1 to present the proposed Cloud-Native LMS architecture and prototype planning. ● Explained the backend design and how the Application Layer would support the system functionalities. ● Recorded feedback and recommendations provided by the lecturer
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	regarding the proposed architecture. • Discussed improvements required before proceeding with further prototype development. • Reviewed the feasibility of the planned backend implementation based on the feedback received.
Individual Contributions (as Backend Developer)	• Elaborated on the architectural role of backend services in managing client requests and facilitating data exchange with the persistence layer. • Provided technical clarity on the utilization of RESTful APIs to ensure seamless communication between frontend modules and backend logic. • Presented the conceptual framework for secure user registration and authentication workflows within the system. • Evaluated the proposed database schema to verify its capacity to accommodate core LMS functionalities and various user permissions. • Refined backend security protocols and storage management strategies following the analytical review and feedback from the supervisor.
Challenges Encountered	Key challenges involved articulating technical backend concepts, addressing data security concerns, balancing system scalability with practical prototype development, and interpreting lecturer feedback to prioritize necessary improvements.
Action Plans	The action plan focuses on refining authentication workflows with enhanced security, optimizing database design, prioritizing feedback from Project Observation 1, developing a clearer implementation plan, and strengthening expertise in backend security and API development.

Table 6 : Weekly Progress Log (Week 9)

Tool Used	Gemini
Purpose	Strengthened technical knowledge regarding backend security protocols and industry standards for safeguarding confidential user data within the LMS.
Prompt Used	"What are the recommended practices for securing user authentication and storing user credentials in a learning management system?"
Output Summary	The system suggested adopting cryptographic password hashing, enforcing role-based access restrictions, and implementing secure session handling while limiting data exposure for private user details.

How It Was Used	The recommendations were used to review the proposed authentication workflow and identify potential improvements to the backend design. The suggested security practices helped me understand how user credentials should be protected and how access permissions could be managed according to user roles. This information was also considered when discussing enhancements to the LMS after receiving feedback during Project Observation 1.
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Table 7: GenAI Usage Record (Week 9)

WEEK 10

Undertaken Tasks & Weekly Activities	- Continued the development planning for the LMS prototype based on feedback received during Project Observation 1. - Reviewed the proposed backend workflows and identified areas that required refinement before implementation. - Discussed how database structures would support the functional requirements of the LMS. - Explored suitable approaches for implementing backend logic and handling data exchange between system components. - Worked with team members to ensure that the backend design remained consistent with the architecture documentation.
Individual Contributions (as Backend Developer)	- Established the definitive data architecture by designating primary entities and relational attributes for the LMS services. - Conducted a technical assessment of CRUD operations supporting User Management, Course Management, and academic tracking modules. - Analyzed backend service logic regarding request validation and data processing protocols prior to persistence layer interaction. - Validated the alignment of planned backend workflows with the functional system specifications established in the project proposal. - Assisted in the formal documentation of backend operational flows to facilitate technical guidance for future implementation phases.
Challenges Encountered	The challenges included determining an optimal database structure, ensuring backend alignment with architectural requirements, proactively identifying potential data flow issues, and managing time effectively during documentation and planning.
Action Plans	The action plan involves refining the database schema, preparing API endpoint specifications, reviewing backend best practices and coordinating team efforts to ensure seamless system integration and implementation.

Table 8 : Weekly Progress Log (Week 10)

Tool Used	ChatGPT
Purpose	Strengthened technical knowledge regarding the conceptualization of a scalable data architecture and the establishment of technical specifications for RESTful API integration within the LMS framework.
Prompt Used	"How should I design a database schema and RESTful API structure for a cloud-native learning management system?"
Output Summary	The system recommended the identification of primary entities including users, courses, assignments, submissions, notifications, and academic records. Furthermore, it offered technical illustrations for RESTful API endpoints and detailed the integration of CRUD operations across the various LMS modules.
How It Was Used	The recommendations were used as a reference when reviewing the proposed database structure and identifying the API operations required for each LMS module. The examples provided helped me visualise how backend services would process requests and interact with the database. This improved the clarity of the backend documentation and provided a better understanding of the implementation approach for the prototype.

Table 9: GenAI Usage Record (Week 10)

WEEK 11

Undertaken Tasks & Weekly Activities	- Continued reviewing the LMS prototype progress to ensure that the proposed backend design was practical and achievable. - Evaluated whether the planned backend workflows supported the functional requirements specified in the project documentation. - Discussed potential testing scenarios for core LMS functionalities such as user authentication and assignment submission. - Reviewed the interaction between backend services and the database layer to identify possible issues before implementation. - Assisted the team in preparing updates on prototype progress and documenting improvements made after Project Observation 1.
Individual Contributions (as Backend Developer)	- Conducted a technical review of the architectural logic governing User Management, Assignment Submission, and Progress Tracking modules. - Validated the alignment of planned CRUD operations with the functional system specifications established for each LMS module. - Assessed the capacity of the database schema to manage complex data relationships and diverse user access permissions.

	• Determined essential validation protocols for backend logic, including the prevention of redundant registrations and enforcement of submission deadlines. • Assisted in the formal documentation of backend testing scenarios and anticipated system responses during operational execution.
Challenges Encountered	Main challenges included technical uncertainty without a test environment, architectural alignment and balancing design with prototype goals.
Action Plans	Merancang pengujian, memperbaiki logik sistem, dan menyediakan dokumentasi backend untuk fasa seterusnya.

Table 10 : Weekly Progress Log (Week 11)

Tool Used	ChatGPT
Purpose	To understand common backend validation techniques and testing considerations for LMS functionalities.
Prompt Used	"What validation rules and testing scenarios should be considered when developing backend functionalities for a learning management system?"
Output Summary	The system output recommended the establishment of rigorous validation protocols, including the mitigation of redundant user registrations, the verification of mandatory input parameters, and the strict enforcement of submission temporal constraints. Furthermore, it emphasized implementing role-based access restrictions and conducting comprehensive scenario testing to validate the operational integrity of the backend framework.
How It Was Used	The recommendations were used to identify the validation rules and expected behaviours that should be considered for the LMS backend. The suggested testing scenarios helped me think about how different user actions, such as invalid logins, duplicate registrations, and late assignment submissions, should be handled by the system. These insights were incorporated into the backend review process to ensure that the proposed workflows aligned with the functional requirements and provided a better user experience.

Table 11: GenAI Usage Record (Week 11)

PHASE 4: Project Observation

WEEK 12

Undertaken	• Participated in Project Observation 2 to present the updated
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Tasks & Weekly Activities	progress of the Cloud-Native LMS project. ● Reviewed the improvements made to the proposed backend design following feedback from Project Observation 1. ● Discussed the readiness of the backend components and database design for the final phase of the project. ● Evaluated whether the planned backend workflows aligned with the functional and non-functional requirements of the LMS. ● Recorded additional feedback and recommendations provided during the observation session for further refinement.
Individual Contributions (as Backend Developer)	● Detailed the architectural refinements made to backend services and persistent storage models following the integration of prior consultative feedback. ● Elucidated the implementation of RESTful API protocols to facilitate standardized data exchange between the presentation layer and system logic modules. ● Conducted a technical verification of the data schema to confirm support for complex entities such as user permissions, task submissions, and academic tracking. ● Provided technical clarity on how validation logic and secure authentication workflows enhance the operational integrity and protection of user credentials. ● Collaborated in the analytical review of the backend framework to isolate and resolve any technical hurdles prior to final documentation completion.
Challenges Encountered	● Explaining the proposed backend improvements clearly despite not having a fully implemented prototype. ● Ensuring that all lecturer feedback from the previous observation had been appropriately addressed. ● Demonstrating how the planned backend processes would meet both functional and non-functional requirements. ● Balancing the level of technical detail during the presentation so that it remained understandable to the audience.
Action Plans	● Finalise backend-related documentation for inclusion in the final report. ● Review all backend workflows and database descriptions to ensure consistency across project deliverables. ● Address any remaining comments received during Project Observation 2 before the final submission. ● Prepare supporting explanations for backend decisions and implementation approaches. ● Conduct a final review of all project documents to ensure accuracy and completeness.

Table 12 : Weekly Progress Log (Week 12)

Tool Used	ChatGPT
Purpose	To obtain suggestions for improving backend documentation and ensuring that the proposed design addressed the project requirements effectively.
Prompt Used	"How can I review and improve backend documentation for a cloud-native learning management system before final submission?"
Output Summary	The system output recommended the verification of structural alignment between functional specifications, architectural visualizations, and backend operational flows. Furthermore, it emphasized providing technical clarity regarding validation protocols, secure authentication workflows, and the definitive responsibilities designated for every microservice.
How It Was Used	The recommendations were used as a checklist to review the backend sections of the project documentation before Project Observation 2. I compared the functional requirements with the proposed backend workflows and database descriptions to identify any inconsistencies. The suggestions also helped me prepare clearer explanations regarding authentication, validation rules, and service responsibilities during the observation session.

Table 13: GenAI Usage Record (Week 12)

WEEK 13

Undertaken Tasks & Weekly Activities	• Participated in the final review and preparation of the project deliverables before submission. • Reviewed the completed project documentation to ensure consistency between the proposal, architecture design, and final report. • Assisted the team in checking that all functional and non-functional requirements had been adequately addressed. • Verified that the backend-related explanations accurately reflected the proposed Cloud-Native LMS architecture. • Conducted a final proofreading session to identify and correct any errors in the documentation.
Individual Contributions (as Backend Developer)	• Conducted a technical validation of the backend report sections, focusing on service definitions, data persistence models, and inter-module communication protocols. • Verified the structural alignment between documented server-side workflows, functional system specifications, and the finalized architectural visualizations.

	• Cross-referenced the persistent storage schema and designated service duties to maintain technical consistency across the entire deliverable. • Refined technical explanations concerning secure authentication logic, academic task processing, and student progress monitoring from an engineering perspective. • Contributed to the final quality assurance audit by isolating and rectifying technical discrepancies or ambiguities prior to project submission.
Challenges Encountered	The main challenges were ensuring document consistency, identifying discrepancies, managing review time, and maintaining clear technical explanations.
Action Plans	The final tasks involved submitting and archiving project documents, reflecting on the backend design, enhancing technical knowledge, and applying lessons learned to future projects.

Table 14 : Weekly Progress Log (Week 13)

Tool Used	ChatGPT
Purpose	To improve the clarity and professionalism of backend-related explanations in the final report.
Prompt Used	"How can I improve the clarity and structure of backend documentation in a software architecture report?"
Output Summary	The system output recommended the verification of structural alignment between functional specifications, architectural visualizations, and backend operational flows. Furthermore, it emphasized providing technical clarity regarding validation protocols, secure authentication workflows, and the definitive responsibilities designated for every microservice.
How It Was Used	The recommendations were used to review and refine the backend sections of the final report before submission. I compared the descriptions of backend services, database interactions, and API communication with the architecture diagrams and functional requirements to ensure consistency. The suggestions also helped improve the clarity of technical explanations, making the documentation easier to understand while maintaining its accuracy and professionalism.

Table 15: GenAI Usage Record (Week 13)